Category Theory for Mathematical Physics — Exercise Sheet 13

Boundaries, defects and non-invertible symmetries

Variant: Questions only

Conventions as in Lecture 13: the toric code has $A_s = \prod X$ on the edges at a vertex and $B_p = \prod Z$ around a face; $e = (0, \chi_-)$, $m = (1, 1)$, $\varepsilon = e \otimes m$ (here ε is the fermion), and its anyons are as in Theorem 12.8. An algebra is (L, μ, ι) and $d(L)$ its dimension. On the Ising chain of n sites, $\eta = \prod_j X_j$, and $\mathcal{T}$ is the translation, $\mathcal{T} O_j \mathcal{T}^{-1} = O_{j+1}$ (not the modular T); D is the Kramers–Wannier operator of § 13.3, and indices are mod n. $\zeta = e^{2\pi i/3}$, $\varphi = (1 + \sqrt{5})/2$, and ω is a 3-cocycle. In Exercise 6, $\varepsilon(b, c)$ is instead Sheet 3's carry, $\zeta_4 = i$, and ψ is a 2-cochain.

Exercise 1 (The toric-code boundaries on the lattice). (i) Draw a piece of the lattice with a smooth boundary on top and a rough one below. On it draw a string of Z's from a bulk vertex s down to the end of a dangling edge, and a string of X's on the edges crossed by a dual path from a bulk face p up through the smooth boundary. Show that each string violates only the term at its bulk end.

(ii) Show that a Z-string from s that ends at a vertex of the smooth boundary violates the star there too, and that an X-string from p that leaves through the rough boundary violates a three-edge plaquette. What does ε, brought to each boundary, leave behind?

(iii) The *dual lattice* has a vertex in each face and an edge across each edge. Show that passing to it, and exchanging X with Z, turns stars into plaquettes and back, the smooth boundary into a rough one, and e into m.

Exercise 2 (Lagrangian algebras). (i) The double semion $\mathcal{Z}(\mathrm{Vec}^{\omega}_{\mathbb{Z}_2})$, $\omega(1,1,1) = -1$, has the four simple objects $(\delta_0, \pm 1)$, $(\delta_1, \pm i)$ (Sheet 12, Exercise 6; spins as in Lecture 12), all invertible of dimension 1, with spins $1, 1, i, -i$ (the self-braidings). Let $b = (\delta_0, -1)$. Show that $b \otimes b \cong 1$, and, as in the proof of Theorem 13.3, that the only Lagrangian algebra is $1 \oplus b$. How does this match Theorem 13.4 and the module categories over $\mathrm{Vec}^{\omega}_{\mathbb{Z}_2}$ (Lecture 11)?

(ii) In $\mathcal{Z}(\mathrm{Fib}) \simeq \mathrm{Fib} \boxtimes \mathrm{Fib}^{\mathrm{rev}}$ the simple objects 1, $\tau \boxtimes 1$, $1 \boxtimes \tau$, $\tau \boxtimes \tau$ have $d = 1, \varphi, \varphi, \varphi^2$ (Sheet 12, Exercise 5). Show that a connected L with $d(L)^2 = \dim \mathcal{Z}(\mathrm{Fib})$ is $1 \oplus (\tau \boxtimes \tau)$ as an object. With Sheet 11, Exercise 4(iii), conclude that $\mathcal{Z}(\mathrm{Fib})$ has exactly one Lagrangian algebra, and, by Theorem 13.5 and the physical principle of Lecture 13, exactly one gapped boundary.

Exercise 3 (Condensing e). Let $L = 1 \oplus e$ in $\mathcal{D} = \mathcal{Z}(\mathrm{Vec}_{\mathbb{Z}_2})$, with the product of Theorem 13.3. A right L-module (M, ρ) is *local* if $\rho \circ c_{L,M} \circ c_{M,L} = \rho$. You may use: every L-module is a direct summand of a free one, $X \otimes L$ with $\rho = \mathrm{id}_X \otimes \mu$ (L is separable); the local modules form a braided fusion category $\mathcal{D}^{\mathrm{loc}}_L$, the phase left after condensing L, with unit L; and, as L is commutative, $X \otimes L$ with X simple is local exactly when the double braiding $c_{L,X} \circ c_{X,L}$ is the identity of $X \otimes L$.

(i) Show that $e \otimes L \cong L$ and $\varepsilon \otimes L \cong m \otimes L$ as L-modules, and that L and $m \otimes L$ are simple modules.

(ii) Show that L is local and $m \otimes L$ is not.

(iii) Conclude that $\mathcal{D}^{\mathrm{loc}}_L \simeq \mathrm{Vect}_{\mathrm{fd}}$: condensing e leaves the trivial phase. What happens to m and ε?

Exercise 4 (The four boundaries of $\mathcal{Z}(\mathrm{Vec}_{S_3})$). Write t for the class of transpositions and u for that of 3-cycles, and the eight anyons of Sheet 12, Exercise 3, as $1_{\mathbf{1}}, 1_{\mathrm{sgn}}, 1_{\mathrm{std}}$ (class $\{1\}$), $t_\pm$ and $u_1, u_\zeta, u_{\zeta^2}$ (the character of the centraliser at the class), with dimensions $1,1,2,3,3,2,2,2$ and spins $1,1,1,1,-1,1,\zeta,\zeta^2$.

(i) Explain why $\mathcal{Z}(\mathrm{Vec}_{S_3})$ has exactly four Lagrangian algebras, and, by the physical principle, four gapped boundaries.

(ii) You may use (Beigi, Shor and Whalen, arXiv:1006.5479, §4.1, eq. (19)) that the boundary built from $\mathrm{Mod}(\mathbb{C}[H])$, $H \leq S_3$, condenses the anyons of the $D(S_3)$-module A_H with basis the pairs (xH, g), $g \in xHx^{-1}$, of flux g, on which $k \in S_3$ acts by $k\cdot(xH,g) = (kxH, kgk^{-1})$; the multiplicity of $([g],\pi)$ in A_H is that of π in the permutation representation of the centraliser of g on the pairs of flux g. Decompose A_H for the four classes of H.

(iii) Check that each A_H has dimension $6 = \sqrt{\dim \mathcal{Z}(\mathrm{Vec}_{S_3})}$ and only summands of spin 1. Which boundary condenses only pure charges, and which only pure fluxes?

Exercise 5 (Kramers–Wannier on the ring). Use the relations $DX_j = Z_{j-1}Z_jD$, $DZ_{j-1}Z_j = X_{j-1}D$ and $\eta D = D$ of § 13.3, on a ring of $n \geq 3$ sites.

(i) Show that $D^2X_j = X_{j-1}D^2$ and $D^2Z_{j-1}Z_j = Z_{j-2}Z_{j-1}D^2$, and that $\mathcal{T}D^2$ commutes with every X_j and every $Z_{j-1}Z_j$.

(ii) Show that an operator commuting with every X_j and every $Z_{j-1}Z_j$ is $a + b\eta$ for numbers a, b.

(iii) Deduce that $D^2 = a(1+\eta)\mathcal{T}^{-1}$ (Seiberg, Seifnashri and Shao normalise D so that $a = 1$). Why is this not the fusion rule $N \otimes N = 1 \oplus \eta$ of a fusion category?

(iv) In *Integrability*, Lecture 18, the chain is $H = -J\sum_k \sigma^x_k\sigma^x_{k+1} - h\sum_k \sigma^z_k$, with parity $\Pi = \prod_k \sigma^z_k$ (its site written k here). Write the Kramers–Wannier relations and D^2 in those letters. Shao uses $D_{\mathrm{Shao}} = D^\dagger/\sqrt{2}$, with D normalised as in (iii), and his T_{Ising} moves O_j to O_{j+1}: compute D^2_{Shao}.

Exercise 6 (Full gauging and the anomaly). (i) For the cocycles ω_p of $\mathbb{Z}_4$ (Sheet 3), $p \in \{0,1,2,3\}$, use the invariant $Q_\omega(1) = \prod_{r=0}^{3}\omega(1,r,1)$, which is unchanged when ω is multiplied by a coboundary, to show that $\mathrm{Vec}^{\omega_p}_{\mathbb{Z}_4}$ is anomalous for $p \neq 0$. Which p allow a rank-one module category over the part on $\{0,2\}$ (Sheet 11, Exercise 2)?

(ii) Gauging all of Vec_{S_3}: using $H^2(S_3, \mathbb{C}^\times) = 0$ (given, as in Sheet 11, Exercise 3), how many choices of ψ are there, and what are the simple lines of the dual symmetry? Show that std has no inverse.

Exercise 7 ⋆. *Ground states on a cylinder.* Put the toric code on a cylinder: circumference $W \geq 2$, vertex rows $y = 0,\dots,\ell$ ($\ell \geq 1$), and a smooth or rough boundary at each end (a rough end has dangling edges and three-edge plaquettes). You may use *Quantum Computing*'s counting: r independent commuting Pauli generators on n qubits, with $-\mathbb{1}$ not in the group they generate, fix a space of dimension 2^{n-r}.

(i) With a rough boundary below and a smooth one on top, count the qubits, stars and plaquettes, and show that the ground space is one-dimensional.

(ii) Do the same with both ends rough, and with both smooth: the ground space has dimension 2.

(iii) Compare with the number of anyons that condense on both boundaries, counted with multiplicity,

$$\dim \mathrm{Hom}(L_{\mathrm{top}}, L_{\mathrm{bottom}}),$$

for $L = 1 \oplus e$ (rough) and $1 \oplus m$ (smooth). Which string operators act on the two ground states when both ends are rough?